



Structured Approach to Advanced Candidate Selection and Treatment Design of Stimulation Treatments

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Abstract

Because many oilfield workers see matrix treatments of wells as a low-tech operation, they often fail to pay attention to candidate selection and treatment design. This lack of attention may have led to the relatively low success rate of these treatments. A 1997 survey in a major oil company indicated that one out of every three to four jobs fails to produce more oil or gas after the treatment. This failure represents a loss to the company of over \$10 million (U.S.), plus a missed extra production capacity of nearly 40,000 BOPD. The probable main cause for this poor performance is the lack of a structured approach to the following:

- selecting the right candidate wells and the appropriate treatment
- defining and implementing a structured treatment design procedure

To improve the situation, a task force investigated the problem and mapped out a total process, which consists of the following steps: A candidate well is selected by comparing its actual performance against its theoretical potential. Then, the source of poor performance is identified, if applicable. Based on this information, the treatment type can be identified and designed, ultimately resulting in an operational stimulation program.

The task force concluded that individual pieces of design software and some design rules existed for many elements, but they lacked an integrated overall approach. The team decided to create a software package by integrating “fuzzy” rules with

appropriate mathematical models to guide field engineers through the individual design steps in a consistent, structured manner. This paper describes various elements of the integrated software package that was designed to meet these needs.

Background

Soon after the first wells were drilled, people started to look for methods to improve the production of new and existing wells. In 1895, a well in Ohio, U.S.A. was successfully treated with hydrochloric acid (HCl) for the first time. However, until the early 1930's, when arsenic inhibitors emerged, the lack of good corrosion inhibitors prevented acid from being widely used to stimulate wells. The use of “mud acid” was then introduced for sandstone wells.

In subsequent years, additional techniques, materials, and equipment were further developed, leading to widespread stimulation activity. Today, several thousand treatments are completed worldwide each year, with a total expenditure of many millions of U.S. dollars. Most of this money is spent on matrix-acidizing treatments.

Stimulation is popular primarily because it is probably the most economical way to generate extra production capacity. For instance, a Dutch oil and gas company carried out a stimulation campaign in the prolific Groningen gas field, although the well rates were already approximately 1.5 million m³/d. A total of 16 wells were treated with acid, resulting in an approximate net increase of 3.6 million m³/d of gas (at 100 bar FTHP) at a total cost of \$600,000 (U.S.). Had the company chosen to obtain such potential by drilling new (infill) wells, they would have spent approximately \$7.5 million.

In spite of this success and potential benefit, the industry did not understand the theoretical aspects of stimulation needed to create sound design methods until the late 1960's. The effort initially concentrated on hydraulic fracturing,^{1,2} but the matrix-acidizing process was also addressed in the early 1970's. Labrid *et al.* published the first comprehensive study on the chemistry of sandstone acidizing in which the authors extensively discussed the largely unknown problem of secondary precipitates.³ Other workers subsequently addressed chemical and physical processes in both sandstone and carbonate acidizing.⁴⁻⁶ Once processes were better understood, new chemical systems were developed.⁷⁻¹² Some of these products are still marketed today,

while others were short-lived because of a continued poor understanding of the processes involved.

The industry also studied the root cause of formation impairment, which is the primary reason matrix treatments were applied.¹²⁻¹⁴ After developing a basic understanding, researchers looked for methods to translate the theory to practical design methods. The first design methods were based on acid response tests on core material.¹⁵ Paccaloni, McLeod *et al.* developed more practical approaches^{16,17} that are based largely on the experiences of oilfield engineers. More theoretically oriented design methods were also developed on the basis of sophisticated, complex chemistry.¹⁸⁻²² At the same time, studies describing the placement and diversion of acid treatments emerged.²³⁻²⁵

Despite the extensive research effort, which probably amounts to hundreds of man-years industry-wide, the failure rate of matrix-acidizing treatments today is estimated at 60 to 70%. Worldwide, this failure rate represents a loss of millions of dollars in wasted stimulation expenditure and missed production. The primary causes for failed treatments have been poor candidate selection and poor treatment design with respect to fluid selection and placement. To spend a stimulation budget most efficiently, the customer needs an overall reservoir solution. By evaluating all the wells in a field and their skin components, an engineer can identify not only acid candidates, but also other stimulation/workover opportunities. The ability to compare the wells, select an appropriate candidate, and analyze all possible solutions for the field (regardless of the type of stimulation/remediation required), results in an Integrated Solutions approach that yields more BOPD/\$ spent on stimulation.

Another reason for the high failure rate of matrix-acidizing treatments is the lack of proper technology transfer to the field. Because the industry often sees matrix treatments as low-tech treatments, little attention is given to design. The technology developed over the years is simply not applied. Information that is not transferred to front line stimulation and production engineers has little value. Bartko *et al.*²⁶ presented one of the first successful attempts to address this problem. Other attempts have also been made to provide an overall design tool,²⁷⁻³⁰ but these tools did not provide the required overall reservoir solution. This paper describes a new tool for fast, complete technology transfer through a user-friendly, easy-to-use browser software program. This program puts the latest technology and a structured, engineered solution at the fingertips of every field engineer.

Design Principles

A well treatment is a logical process that requires a number of ordered steps before the desired results can be achieved. This process enables the production technologist or stimulation engineer in the field to select and design the best suitable treatment to improve the productivity of a damaged well.

The structured process of a stimulation job (**Fig. 1**, Page 7) consists of the following phases:

1. **Candidate Selection and Damage Identification**—In this phase, the best candidate well for stimulation treatment is selected, regardless of the type of stimulation/remediation needed. During this phase, the best treatment for the type of damage is also determined.
2. **Fluid Selection**—In this phase, the appropriate fluids, volumes, and additives are selected.
3. **Placement**—This phase focuses on placing the matrix acid treatment, including diversion, preparation of a pumping schedule with volumes, rates, etc., and simulation of the treatment.
4. **Treatment Evaluation**—In this phase, the actual results of a real stimulation treatment are evaluated for changes in skin compared to the expected and predicted results.

These phases were used as a basis for developing the new software program.

Phase 1—Candidate Selection and Damage Identification.

Candidate Selection. Production may decline for numerous reasons. This decline may be caused by the natural characteristics of the fluids or rock being produced (fines, wax, etc.), by wellbore damage from the drilling/completion process (such as mud remnants), or by mechanical difficulties in the overall completion process. The natural production may also be low because of disappointing reservoir properties, such as low permeability. All of these problems will result in additional pressure drop and thus affect the skin term.

The skin factor “S” is a dimensionless, mathematical concept for the ease with which fluids flow from the undisturbed reservoir to the wellbore. It represents the pressure drop caused by additional resistance to the flow of reservoir fluid in the near-wellbore region. The value is the combined effect of several parameters, including formation damage. To properly interpret the skin and therefore determine the appropriate remedial action, field engineers must analyze the contribution of each factor. This analysis may result in additional opportunities for production improvement such as reperforating, but only those candidates with a damaged skin will benefit from stimulation/remediation of the formation itself. Therefore, the key to candidate selection lies in the analysis of the various skins.

In this module, the ideal production of a well refers to the expected production based on general information about the (undamaged) reservoir quality, such as permeability, thickness, porosity, saturation, etc. Several models can be used to calculate this potential production, ranging from the simple application of Darcy’s law, to the use of more complicated reservoir simulators. Skin or WIQI (flow efficiency) data can also be applied.

The skin factor is frequently determined with a Horner plot of the pressure data from a pressure buildup test. For the purposes of candidate selection, the following skin components, as derived by various authors,³¹⁻³⁵ are evaluated and provided

after the following equation. Detailed definitions of each skin term are provided in **Table 1** (Page 7).

The real skin caused by damage (the portion of the total skin that can be removed by matrix treatments) can be isolated from the total skin when the following formula is used:

$$S_{dam} = S_{tot} - (S_{perf} + S_{turb} + S_{dev} + S_{gravel} + S_{perf\ size}) \dots (1)$$

Where

S_{tot}	=	total skin factor (“Horner”skin)
S_{dam}	=	skin resulting from formation damage
S_{perf}	=	skin resulting from limited perforation height
S_{turb}	=	skin resulting from turbulent (non-Darcy) flow
S_{dev}	=	skin resulting from wellbore deviation
S_{gravel}	=	skin resulting from gravel packing
$S_{perf\ size}$	=	skin resulting from a small perforation

Basically, during the candidate selection process, the field engineer compares several wells based on criteria such as the potential for improved productivity, damaged skin, flow efficiency, or other parameters, and ranks the ones that deserve further research, possibly resulting in an acid treatment design, a fracturing design, or a workover. One goal of the software is to help users wisely apply their stimulation resources (help them treat the right wells and *avoid* wasting money on the wrong ones). Upon performing this phase of the process, the field engineer may determine that the total solution is three acid treatments, two fracturing jobs, one conformance treatment and three reperforating runs—not a matrix-acidizing campaign on 12 wells.

Damage Identification. If a well scores high in the damage skin column and appears to suffer from near-wellbore damage (a matrix-treatment candidate), the user can continue the selection procedure by trying to classify the nature of the damage. In principle, formation damage is classified according to the process or operation that caused the damage to develop. Damage mechanisms that should be considered include the following:^{36,}

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- scales
- fluosilicates
- precipitated silica
- asphaltene
- paraffin
- production chemicals
- bacteria problems
- clay swelling
- clay migration
- fines migration
- solids/plugging
- mud damage
- emulsion block

- polymer damage
- salt bridge
- oil carryover
- water block
- wettability change
- not damaged

The type and depth of the damage impacts the type of treatment(s) that will be the most effective for each well. A haphazard approach to treatment selection without considering the cause of the formation damage will result in fewer “successful” treatments.

Phase 2—Fluid Selection. The next phase of this design approach focuses on fluid selection. The software gives users three options regarding fluid design:

- an expert system
- a geochemical simulator
- user-specified information

Users can apply these methods independently or in combination to obtain a complete fluid series.

Expert System. The expert system uses logical rules based on engineering fundamentals, the latest laboratory research,¹⁹⁻²² and relationships determined through experience, guidelines, and best practices for treatment design. This method generates a full suite of fluid systems, including acid selections, conditioner selections, volumes, and additives for both sandstones and carbonates.

Geochemical Simulator. The geochemical simulator performs an iterative, matrix-driven geochemical simulation based on the acid fluid type and formation mineralogy. It is a fundamentally more rigorous, theoretical calculation based on physics, chemistry, and thermodynamics. This method simulates acid invading the rock matrix and determines the optimum compromise between the acid’s power to dissolve the clay components and the potential of reaction products precipitating and impairing the results. It also considers how acid volume could affect the loss of formation integrity should excessive quantities of minerals be dissolved during the procedure.

Phase 3—Placement. Once users have determined the type of near-wellbore damage and have established the most effective acid composition for damage removal, they must design a treatment that will effectively place the fluid so that it will remove the damage hindering production. Therefore, the placement is as important as the fluids selected. Placement includes (1) diversion, (2) various placement techniques (3) the determination of the complete pumping schedule with stages, volumes,

and rates, and (4) the simulation of the fluid placement to optimize the design process.

Diversion methods that can be designed and simulated with this approach include ball sealers, degradable particulates, foams, gel slugs, and gelled acids. Other placement techniques such as maximum rate-maximum pressure pumping (MAPDIR)¹⁶ and coiled tubing (CT) with movement can also be designed and simulated. In addition, because the treatment interval can be designed across any part of the identified reservoir layers, mechanical isolation techniques such as packers/bridge plugs and selective injection packers can be evaluated.

Once the user has determined the fluids and diversion techniques, the new system will automatically generate a pumping schedule. This schedule includes the stages and quantities of fluid, identifies the diversion stages, and indicates the volume/number of diverters, the rates down applicable flow paths, and the gas volumes (if applicable). The user can then export this schedule as a report or optimize/evaluate it with the simulator.

The fluid placement simulator simulates pumping a treatment into the well and can be a valuable tool for treatment design and analysis. The current version of the simulator manages the following variables:

- a multiple-stage treatment with both Newtonian and non-Newtonian fluids
- multiple-interval formation with skin
- sandstones (HF-acid) and carbonates (wormholing)
- openhole completions, with or without gravel pack
- bullheading, (moving) CT, and simultaneous pumping through the annulus and CT
- tubing friction
- particle, viscous fluid, and foam diversion
- ball-sealer diversion

The simulator also allows users to answer such questions as the following:

- Where do the fluids go when pumped downhole?
- Which intervals take the most fluid, and which take the least fluid?
- How deep does the acid penetrate into the formation? What is the skin reduction?
- Is diversion or coiled tubing needed for improved fluid coverage?
- At what rate can we pump the job? Is there excessive friction?
- What must the rate be to ensure efficient wormholing in a carbonate?

Sometimes users may find it helpful to compare the results of different simulator runs. With this software, they can simulta-

neously perform simulations in two different windows. For example, users can view two dynamic fluid placement plots simultaneously in two separate windows. This feature is particularly useful when users are trying to determine the effects of diversion.

Phase 4—Treatment Evaluation. The final phase in the structured approach is to evaluate the results and enter this information into the design system. While mathematically simulating the treatment, users can only predict the bottomhole skin profile for the job on the basis of time. After performing the treatment, users can import actual job data, generate another skin profile, and compare the two results. They can also evaluate the “real job” in the fluid simulator. This evaluation completes the process and allows users to refine the structured approach continually.

Horizontal Wells. The same structured approach is also used for horizontal wells. Users can calculate the productivity parameters in a horizontal well using equations that are markedly different than those used for vertical/deviated wells. For example, in a given reservoir at similar actual production rates, a skin evaluation would typically show that the drawdown pressure for a horizontal well is dramatically lower than that of a vertical well.

The key graphics and calculations needed for accurate production estimates and treatment designs for horizontal wells are based on the proper definition of well trajectory, including pay intervals. Allowances are also made if the reservoir is bounded by an active (pressure-supportive) gas cap and/or a bottom aquifer. Indirect allowances are made for horizontal wellbores in the simulator, several of the advisors, and in fluid volume recommendations, which are adjusted based on the net pay length.

Usability/System Design Considerations

Browser Design. The main screen in the normal operating mode for the software is arranged in three windows with two toolbars and a status bar. (Fig. 2, Page 8). The “tree” or browser window, which provides for easy navigation, contains all of the wells and treatments for that database. The main work area is where users enter data, view help information, and run calculations. The message window at the bottom of the screen provides useful information, error messages, warnings, etc.

The screen contains the main toolbar, along with the standard New, Open, Save, Back, Forward, and Help buttons. Users select the Help button to access context and screen-sensitive help. The second (browser) toolbar contains shortcut buttons for creating new well clusters, and inserting new wells and treatments etc.

One-Point Data Entry. Because the program pieces are fully integrated, users must only enter information once for it to appear in other parts of the program. This feature is possible because the software was developed with “shared data” layer. This data layer will eventually allow programmers to add other

applications (such as water control, sand control, etc.), and communication to corporate and public databases.

Advisors. Advisors are designed to address scenarios in which the user does not know (or will have to use a calculator to determine) a reasonable answer. Specifically, they allow users to (1) access additional help in making decisions, (2) calculate inputs, or (3) play “what-if” scenarios on the effects of changing one parameter across a given range. For example, users can use the diversion advisor to determine the best diversion method for a specific completion/well or to determine the effect that wellbore deviation has on the various types of diversion.

Advisors are present for several functions, but the following are four of the most useful:

- inflow calculator
- redbook
- formation damage advisor
- diversion advisor

Inflow Calculator. This basic tool provides quick, simple estimates of potential production increase with a reduced skin. The evaluation is based on several factors, one of which is the potential ideal flow rate that could be achieved if the skin factor were reduced to zero. This calculation uses the pseudosteady-state form of the Darcy radial flow equation, and assumes that the total skin is reduced to zero for use in comparing the subject well to others.

Redbook. This module is taken from the Cementing Table Redbook for determining pipe OD, ID, weight, grade, etc. Access to this module from within the program is associated with the wellbore configuration and completion information.

Formation Damage Advisor. This tool allows users to rank potential damage mechanisms based on various inputs about the well’s history, completion, production, etc. using a weighted cause-and-effect relationship.

Diversion Advisor. This tool helps users select the appropriate diversion method based on various inputs about the well’s formation characteristics, completion, and production using a weighted cause-and-effect relationship.

Cluster Organization. Most current design software is built for specific single-well applications. To maximize the total reservoir solution, however, users must have access to multiple-well data. This new software is built on a data model that will store multiple-well data. The wells and their treatments can also be arranged with multiple-cluster views so that once entered, users can group wells geographically, geologically, by operator, etc. (**Fig. 3**, Page 8).

Automatic Report Generation. The software has a versatile, user-friendly report generator. Users can extensively customize

reports into specific formats or templates that they can save and use again later. The report is created in Microsoft Word, where it can be further customized as needed. The Report Arrangement tree is a section-by-section, page-by-page outline of what the user’s report will look like when it is created.

The only report pages available for printing are those for which data is available. Users can drag and drop sections and specific pages to customize the report format. Once users have determined the appropriate design, they can generate customer-ready, printed output in minutes.

Conclusions

- High failure rates for matrix-acidizing treatments are usually attributed to poor candidate selection and/or inadequate treatment design.
- Stimulation budgets can be more effectively spent when decisions are based on an overall reservoir solution that considers all types of treatments.
- Because acidizing-fluid selection and placement involve complicated chemistry and technology, information transfer and design are best accomplished with a software design tool.
- A structured approach to appropriate candidate selection and treatment design should increase the success rate and overall production of candidate wells.

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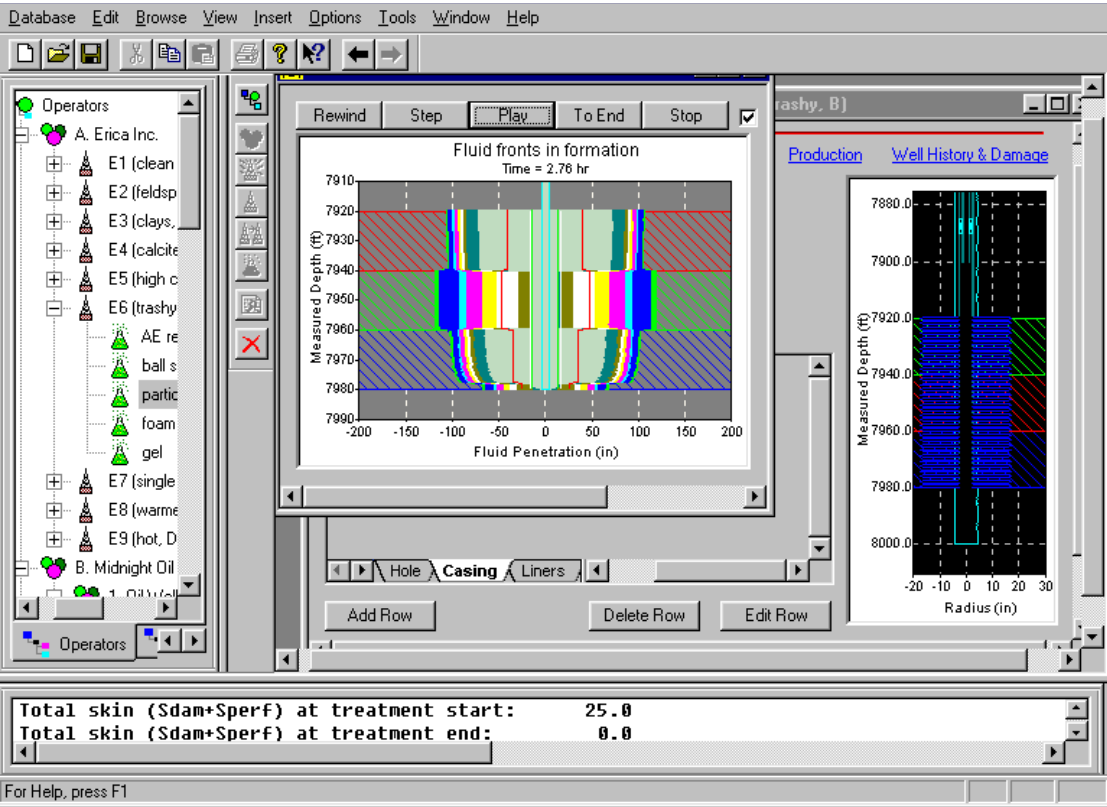


Fig. 2—Main screen for the new software.

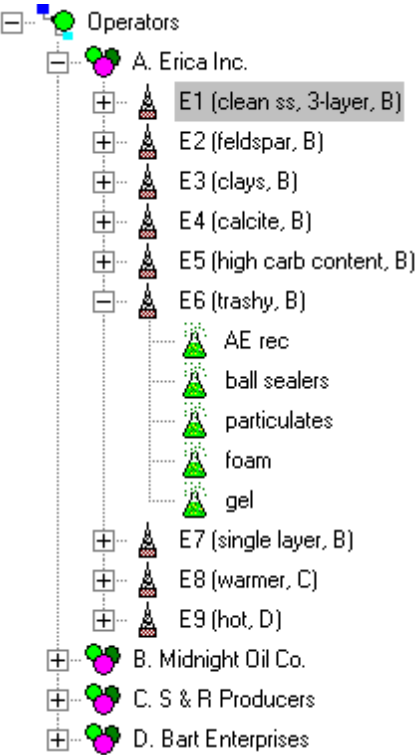


Fig. 3—Data organization for the software.